with an additional space dimension there would be additional warps and ripples. And those warps and ripples, his analysis showed, would be just right to describe electromagnetic fields. In Kaluza's hands, Einstein's own geometrical approach to the universe proved powerful enough to unite gravity and electromagnetism.

Of course, there was still a problem. Although the mathematics worked, there was—and still is—no evidence of a spatial dimension beyond the three we all know about. So was Kaluza's discovery a mere curiosity, or was it somehow relevant to our universe? Kaluza had a powerful trust in theory—he had, for example, learned to swim by studying a treatise on swimming and then diving into the sea—but the idea of an invisible space dimension, no matter how compelling the theory, still sounded outrageous. Then, in 1926, the Swedish physicist Oskar Klein injected a new twist into Kaluza's idea, one that suggested where the extra dimension might be hiding.

The Hidden Dimensions

To understand Klein's idea, picture Philippe Petit walking on a long, rubber-coated tightrope stretched between Mount Everest and Lhotse. Viewed from a distance of many miles, as in Figure 12.5, the tightrope appears to be a one-dimensional object like a line—an object that has extension only along its length. If we were told that a tiny worm was slithering along the tightrope in front of Philippe, we'd wildly cheer it on because it needs to stay ahead of Philippe's step to avoid disaster. Of course, with a moment's reflection we all realize that there is more to the surface of the tightrope than the left/right dimension we can directly perceive. Although difficult to see with the naked eye from a great distance, the surface of the tightrope has a second dimension: the clockwise/counterclockwise dimension that is "wrapped" around it. With the aid of a modest telescope, this circular dimension becomes visible and we see that the worm can move not only in the long, unfurled left/right direction but also in the short, "curled-up" clockwise/counterclockwise direction. That is, at every point on the tightrope, the worm has two independent directions in which it can move (that's what we mean when we say the tightrope's surface is two-